Using Knowledge Elicitation to Improve Web Effort Estimation:

Lessons from Six Industrial Case Studies

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***Abstract*—This paper details our experience building and validating six different expert-based Web effort estimation models for ICT companies in New Zealand and Brazil. All models were created using Bayesian networks, via eliciting knowledge from domain experts, and validated using data from past finished projects. Post-mortem interviews with the participating companies showed that they found the entire process extremely beneficial and worthwhile, and that all the models created remained in use by those companies.**

***Keywords;* Web/software effort estimation; expert-based models; bayesian networks; project management**

I. INTRODUCTION

A cornerstone of Web project management is effort estimation, the process by which effort is forecasted and used as basis to predict costs and allocate resources effectively, so enabling projects to be delivered on time and within budget. Effort estimation is a very complex domain where the relationship between factors is non-deterministic and has an inherently uncertain nature. E.g. assuming there is a relationship between development effort and an application's size (e.g. number of Web pages, functionality), it is not necessarily true that increased effort will lead to larger size. However, as effort increases so does the probability of larger size. Effort estimation is a complex domain where corresponding decisions and predictions require reasoning with uncertainty.

Within the context of Web effort estimation, numerous studies investigated the use of effort prediction techniques. However, to date, only Mendes [9],[10],[11],[12],[13],[16] investigated the explicit inclusion, and use, of uncertainty, inherent to effort estimation, into models for Web effort estimation. Mendes [9],[10],[11] built a Hybrid Bayesian Network (BN) model (structure expert-driven and probabilities data-driven), which presented significantly superior predictions than the mean- and median-based effort [10], multivariate regression [9],[10],[11], case-based reasoning and classification and regression trees [11]. Mendes [12], and Mendes and Mosley [16] extended their previous work by building respectively four and eight BN models (combinations of Hybrid and data-driven). These models were not optimised, as previously done in Mendes [9],[10],[11], which might have been the reason why they presented significantly worse accuracy than regression-based models. Finally, Mendes et al. [19], Mendes [13], Mendes [14], and Mendes [15] each describe a case study where an expert-based Web effort estimation BN model was successfully built and used to estimate effort for projects developed by Web companies in Auckland, New Zealand. This paper combines the experience and findings resulting from the four case studies abovementioned, plus another two (yet to be published), details the process employed to build and validate the BN models, and discusses lessons learned.

A BN is a model that supports reasoning with uncertainty due to the way in which it incorporates existing complex domain knowledge [7]. It represents knowledge using two parts. The first, the qualitative part, represents the structure of a BN as depicted by a directed acyclic graph (digraph) (see Figure 1). The digraph's nodes represent the relevant variables (factors) in the domain being modelled, which can be of different types (e.g. observable or latent, categorical). The digraph's arcs represent the causal relationships between variables, where relationships are quantified probabilistically. The second, the quantitative part, associates a node conditional probability table (CPT) to each node, its probability distribution. A parent node's CPT describes the relative probability of each state (value); a child node's CPT describes the relative probability of each state conditional on every combination of states of its parents (e.g. in Figure 1, the relative probability of Total effort (TE) being 'Low' conditional on Size (new Web pages) (SNWP) being 'Low' is 0.8). Each column in a CPT represents a conditional probability distribution and therefore its values sum up to 1 [7]. Once a BN is specified, evidence (e.g. values) can be entered into any node, and probabilities for the remaining nodes are automatically calculated using Bayes' rule [21]. Therefore BNs can be used for different types of reasoning, such as predictive and "what-if" analyses to investigate the impact that changes on some nodes have on others [6].

Within the context of Web effort estimation there are issues with building data-driven or hybrid Bayesian models, as follows:

1. Any dataset used to build a BN model should be large enough to provide sufficient data capturing all (or most) relevant combinations of states amongst variables such that probabilities can be learnt from data, rather than elicited manually. Under such circumstances, it is very unlikely that the dataset would contain project data volunteered by only a single company (single-company dataset). As far as we know, the largest dataset of Web

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projects available is the Tukutuku dataset (195 projects) [18]. This dataset has been used to build data-driven and hybrid BN models; however results have not been encouraging overall, and we believe one of the reasons is due to the small size of this dataset.

2. Even when a large dataset is available, the next issue relates to the set of variables part of the dataset. It is unlikely that the variables identified represent all the factors within a given domain (e.g. Web effort estimation) that are important for companies that are to use the data-driven or hybrid model created using this dataset. This was the case with the Tukutuku dataset, even though the selection of which variables to use had been informed by two surveys [18]. However, one could argue that if the model being created is hybrid, then new variables (factors) can be added to, and existing variables can be removed from the model. The problem is that every new variable added to the model represents a set of probabilities that need to be elicited from scratch, which may be a hugely time consuming task.

3. Different structure and probability learning algorithms can lead to different prediction accuracy [16]; therefore one may need to use different models and compare their accuracy, which may also be a very time consuming task.

4. When using a hybrid model, the BN's structure should ideally be jointly elicited by more than one domain expert, preferably from more than one company, otherwise the model built may not be general enough to cater for a wide range of companies [16]. There are situations, however, where it is not feasible to have several experts from different companies cooperatively working on a single BN structure. One such situation is when the companies involved are all consulting companies potentially sharing the same market. This was the case within the context of this research.

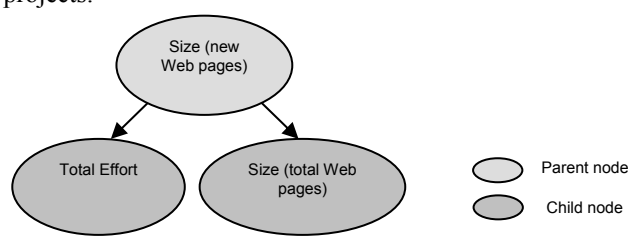
5. Ideally the probabilities used by the data-driven or hybrid models should be revisited by at least one domain expert, once they have been automatically learnt using the learning algorithms available in BN tools. However, depending on the complexity of the BN model, this may represent having to check thousands of probabilities, which may not be feasible. One way to alleviate this problem is to add additional factors to the BN model in order to reduce the number of causal relationships reaching child nodes; however, all probabilities for the additional factors would still need to be elicited from domain experts.

6. The choice of variable discretisation, structure learning algorithms, parameter estimation algorithms, and the number of categories used in the discretisation all affect the accuracy of the results and there are no clear-cut guidelines on what would be the best choice to employ. It may simply be dependent on the dataset being used, the amount of data available, and trial and error to find the best solution [16].

Therefore, given the abovementioned constraints, as part of two government-funded projects on using Bayesian Networks to Web effort estimation (New Zealand and Brazilian governments), several expert-based company-specific Web effort BN models were built and validated, with the participation of five Web companies in New Zealand, and one company in Brazil. The development and successful deployment of these six models is the subject and contribution of this paper.

Note that we are not suggesting that data-driven and hybrid BN models should not be used. On the contrary, they have been successfully employed in numerous domains [25]; however the specific domain context of this paper – that of Web effort estimation, provides other challenges (described above) that lead to the development of solely expert-driven BN models.

We would also like to point out that we have explicitly emphasised that the models focus on Web effort estimation because, in our view, Web and software development differ in a number of areas, such as: Application Characteristics, Primary Technologies Used, Approach to Quality Delivered, Development Process Drivers, Availability of the Application, Customers (Stakeholders), Update Rate (Maintenance Cycles), People Involved in Development, Architecture and Network, Disciplines Involved, Legal, Social, and Ethical Issues, and Information Structuring and Design. A detailed discussion on this issue is provided in [17]. However, despite the differences between Web and software development, we believe that our contribution goes beyond the area of Web engineering given that the process presented herein can also be used to build BN models for any IT company that estimates development effort for their projects.



CPT for node Size (new Web pages)	
Low	0.2
Medium	0.3
High	0.5

CPT for node Total Effort (TE)			
Size (new Web pages)	*Low*	*Medium*	*High*
Low	0.8	0.2	0.1
Medium	0.1	0.6	0.2
High	0.1	0.2	0.7

Figure 1. Example of a BN model and two CPTs.

The remainder of the paper is organised as follows: Section 2 provides an overview of the process used to build and validate the BNs presented herein; Section 3 details this process by concentrating on the six expert-based Web Effort BNs focus of this paper. Common patterns amongst the models are discussed in Section 4, followed by a description of lessons learnt in Section 5. Finally, conclusions and comments on future work are given in Section 6.

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II. General Process Employed to Build BNs

The BNs focus of this research were built and validated using an adaptation of the Knowledge Engineering of Bayesian Networks (KEBN) process proposed in [25] (see Figure 2). Within the context of this work the author was the Knowledge Engineer (KE), and Web project managers from several well-established Web companies in either New Zealand or Brazil were the Domain Experts (DEs).

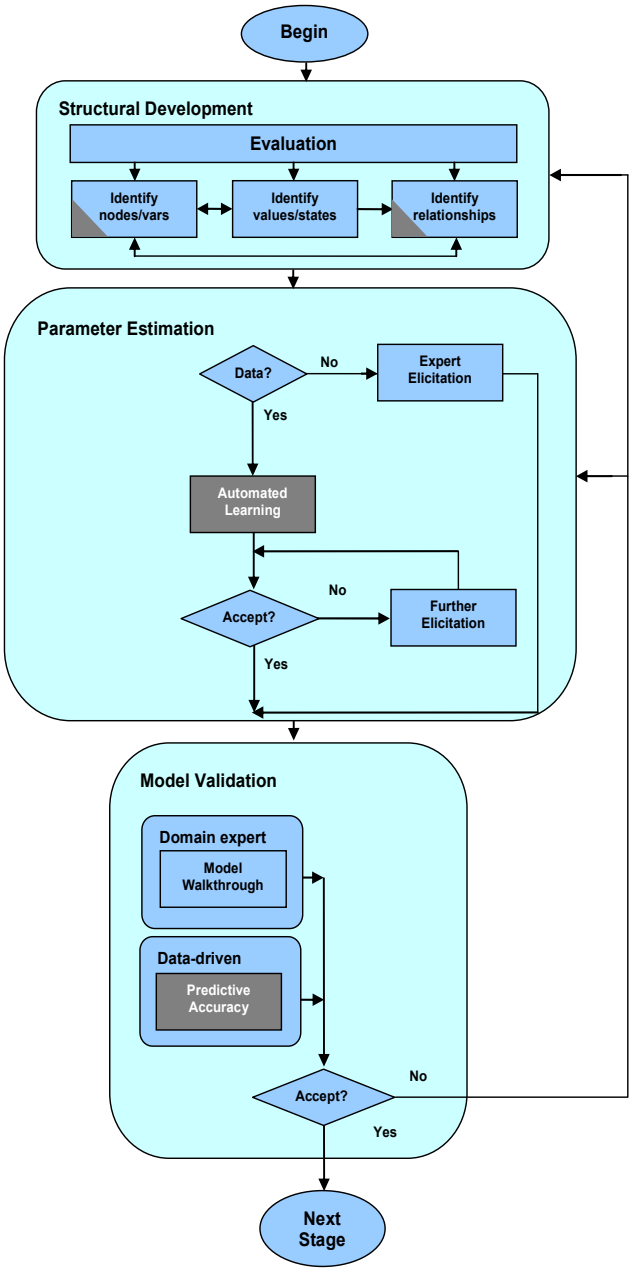

Figure 2. KEBN, adapted from [25]

The three main steps within the adapted KEBN process are the Structural Development, Parameter Estimation, and Model Validation. This process iterates over these steps until a complete BN is built and validated. Each of these three steps is detailed next, and presented in Figure 2:

Structural Development: This step represents the qualitative component of a BN, which results in a graphical structure comprised of, in our case, the factors (nodes, variables) and causal relationships identified as fundamental for effort estimation of Web projects. In addition to identifying variables, their types (e.g. query variable, evidence variable) and causal relationships, this step also comprises the identification of the states (values) that each variable should take, and if they are discrete or continuous. In practice, currently available BN tools require that continuous variables be discretised by converting them into multinomial variables, also the case with the BN software used in this study. The BN's structure is refined through an iterative process. This structure construction process has been validated in previous studies [5],[6],[8],[20],[25] and uses the principles of problem solving employed in data modelling and software development [23]. As will be detailed later, existing literature in Web effort estimation, and knowledge from the DEs were employed to elicit the Web effort BNs' structures. Throughout this step the KE also evaluates the structure of the BN, done in two stages. The first entails checking whether: variables and their values have a clear meaning; all relevant variables have been included; variables are named conveniently; all states are appropriate (exhaustive and exclusive); a check for any states that can be combined. The second stage entails reviewing the BN's graph structure (causal structure) to ensure that any identified d-separation dependencies comply with the types of variables used and causality assumptions. D-separation dependencies are used to identify variables influenced by evidence coming from other variables in the BN [7],[21]. Once the BN structure is assumed to be close to final the KE may still need to optimise this structure to reduce the number of probabilities that need to be elicited or learnt for the network. If optimisation is needed, techniques that change the causal structure (e.g. divorcing [7]) are employed.

Parameter Estimation: This step represents the quantitative component of a BN, where conditional probabilities corresponding to the quantification of the relationships between variables [7],[21] are obtained. Such probabilities can be attained via Expert Elicitation, automatically from data, from existing literature, or using a combination of these. When probabilities are elicited from scratch, or even if they only need to be revisited, this step can be very time consuming. In order to minimise the number of probabilities to be elicited some techniques have been proposed in the literature [4],[5],[24]. In addition, we have also recently proposed a technique to reduce the time needed for probability elicitation, to be discussed later.

Model Validation: This step validates the BN resulting from the two previous steps, and determines whether it is necessary to re-visit any of those steps. Two different validation methods are generally used - Model Walkthrough and Predictive Accuracy. Model walkthrough represents the use of real case scenarios that are prepared and used by DEs to assess if the predictions provided by a BN correspond to

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the predictions experts would have chosen based on their own expertise. Success is measured as the frequency with which the BN's predicted value for a target variable (e.g. quality, effort) that has the highest probability corresponds to the experts' own assessment.

Predictive Accuracy uses past data (e.g. past project data), rather than scenarios, to obtain predictions. Data (evidence) is entered on the BN model, and success is measured as the frequency with which the BN's predicted value for a target variable (e.g. quality, effort) showing the highest probability corresponds to the actual value from past data.

III. Process Used to build the Expert-Based BNs

This sub-Section revisits the adapted KEBN process (see Figure 2), detailing the tasks carried out for each of the three main steps that form part of that process. Before starting the elicitation of the Web effort BN models, the Domain Experts (DEs) from all participating Web companies were presented with an overview of Bayesian Network models, and examples of "what-if" scenarios, using a made-up BN. This, we believe, facilitated the entire process as the use of an example, and the brief explanation of each of the steps in the KEBN process, provided a concrete understanding of what to expect. We also made it clear that the KE was a facilitator of the process, and that the Web companies' commitment was paramount for the success of the collaboration. The effort required by each company to have their BN models created and the characteristics of each model are detailed in Table I.

TABLE I. Characteristics BNs and Number of DEs

	Companies (Country: New Zealand(NZ) or Brazil (BR))					
Characteristics	A (NZ)	B (NZ)	C (NZ)	D (NZ)	E (NZ)	F (BR)
Number of DEs	1	1	2	2	7/2	1
Number of Employees	~5	~5	~20	~30	~100	~30
Number of 3-hours elicitation sessions	12	6	8	12	12/12	20
Total hours to elicit & validate model	36	18	24	36	98	60
Effort to elicit & validate model (person/hours)	72	36	72	108	324	120
Number of factors	14	13	34	33	38	19
Number of relationships	18	12	41	60	50	37
Number of past projects used as validation set	22	8	11	22	22	9

The DEs who took part in the case studies were all project managers of well-established Web companies in either Auckland (New Zealand), or Rio de Janeiro (Brazil), each with at least 10 years of experience in project management. These companies varied in their size, measured as the total number of employees. In addition, all six companies were consulting companies and as such, developed a wide range of Web applications, from static & multimedia-like to very large e-commerce solutions. All six companies employed a wide range of Web technologies, thus also enabling the development of Web 2.0 and Web 3.0 applications. Finally, when approached, they were all looking at improving their current effort estimates, and agreed to participate for two main reasons: i) because the models being created were single-company models geared towards their specific needs; ii) and also because their expertise and participation were acknowledged as essential to eliciting the models.

Detailed Structural Development and Parameter Estimation: In order to identify the fundamental factors that the DEs took into account when preparing a project quote we used the set of variables from the Tukutuku dataset [18] as a starting point (see Table II). We first sketched them out on a white board, each one inside an oval shape, and then explained what each one meant within the context of the Tukutuku project. Our previous experience eliciting BNs in other domains (e.g. ecology, resource estimation) suggested that it was best to start with a few factors (even if they were not to be reused by the DE), rather than to use a "blank canvas" as a starting point [19].

TABLE II. Tukutuku Variables

	Variable Name	Description
Project Data	*TypeProj*	Type of project (new or enhancement).
	nLang	Number of different development languages used
	DocProc	If project followed defined and documented process.
	ProImpr	If project team involved in a process improvement programme.
	Metrics	If project team part of a software metrics programme.
	DevTeam	Size of a project's development team.
	TeamExp	Average team experience with the development language(s) employed.
Web application	*TotWP*	Total number of Web pages (new and reused).
	NewWP	Total number of new Web pages.
	TotImg	Total number of images (new and reused).
	NewImg	Total number of new images created.
	Num_Fots	Number of features reused without any adaptation.
	HFotsA	Number of reused high-effort features/functions adapted.
	Hnew	Number of new high-effort features/functions.
	TotHigh	Total number of high-effort features/functions
	Num_FotsA	Number of reused low-effort features adapted.
	New	Number of new low-effort features/functions.
	TotNHigh	Total number of low-effort features/functions

Within the context of the Tukutuku project, based on collected data, a new high-effort feature/function and a high-effort adapted feature/function require respectively at least 15 and 4 hours to be developed by one experienced developer.

Once the Tukutuku variables had been sketched out and explained, the next step was to remove all variables that were not relevant for the DEs, followed by adding to the white board any additional variables (factors) suggested by them. This entire process was documented using digital voice recorders and also text editors. We also documented

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descriptions and rationale for each factor proposed by the DEs. The factors proposed were indeed influenced by DEs' hunches and insights; however DEs decisions and choices were also very much influenced by their solid previous experience managing Web projects, and estimating development effort.

Next, we identified the possible states that each factor would take. All states were discrete. Whenever a factor represented a measure of effort (e.g. Total effort), we also documented the effort range corresponding to each state, to avoid any future ambiguity. For example, to one of the participating Web companies, 'very low' Total effort corresponded to 4+ to 10 person hours, etc. Once all states were identified and thoroughly documented, it was time to elicit the cause and effect relationships. As a starting point to this task we used a simple medical example from [25] (see Figure 3).

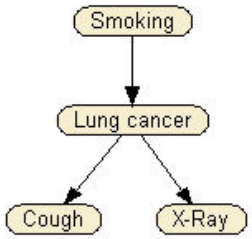

Figure 3. An Example of a Cause and Effect relationship

This example clearly introduces one of the most important points to consider when identifying cause and effect relationships – timeline of events. If smoking is to be a cause of lung cancer, it is important that the cause precedes the effect. This may sound obvious with regard to the example used; however, it is our view that the use of this simple example significantly helped the DEs understand the notion of cause and effect, and how this related to Web effort estimation and the BNs being elicited. Once the cause and effect relationships were identified, we worked on the elicitation of probabilities to quantify each of the cause and effect relationships previously identified. In all four cases, there was an iterative process between the structural development and parameter elicitation steps.

Detailed Model Validation: Both Model walkthrough and Predictive accuracy were used to validate all six Web Effort BN models, where the former was the first type of validation to be employed in all cases. DEs used different scenarios to check whether the node Total_effort would provide the highest probability to the effort state that corresponded to the DE's own suggestion. However, it was also necessary to use data from past projects, for which total effort was known, in order to check the model's calibration. Table I details the number of projects used by each company as validation set. In all cases, DEs were asked to use as validation set a range of projects presenting different sizes and levels of complexity, and being representative of the types of projects developed by their Web company.

For each project in a validation set, evidence was entered in the BN model, and the effort range corresponding to the highest probability provided for 'Total Effort' was compared to that project's actual effort. Whenever actual effort did not fall within the effort range associated with the category with the highest probability, there was a mismatch; this meant that some probabilities needed to be adjusted. In order to know which nodes to target first we used a Sensitivity Analysis report, which provided the effect of each parent node upon a given query node. Within our context, the query node was 'Total Effort'.

Whenever probabilities were adjusted, we re-entered the evidence for each of the projects in the validation set that had already been used in the validation step to ensure that the calibration already carried out had not been affected. This was done to ensure that each calibration would always be an improvement upon the previous one. Once all projects were used to calibrate a model, the DE(s) assumed that the Validation step was complete.

Each of the five New Zealand BNs has been in production for at least 18 months, and the Brazilian BN has been in production since May 2011. Due to shortage of space we cannot show all six BNs; however, Figures 4 and 5 show respectively one of the BN models developed, and the same model with some evidence selected. This model is identified in Table I under Company C.

IV. Common Patterns

All six models were BNs targeting at the estimation of effort for new projects; therefore, we also looked at combining part of results from the six different case studies focusing at two specific points: i) to identify the total set of factors selected by the Web companies; ii) to identify the amount of overlap between companies, measured using a vote counting approach. In order to identify the total set of factors using a common terminology, we employed a methodology that we have previously proposed, which uses the contextual meaning of each factor in order to match factors across different BN models. The details relating to this methodology are outside the scope of this paper; however for those interested, they are documented in [1].

Apart from total effort, which was identified by all participating companies, there were three factors that were chosen by five of the six Web companies:

- Average Project Team Experience with Technology
- Effort to Program Features
- Project Management Effort

All five BN models showed these factors affecting total effort. This includes project management, shown to be affected by other factors such as the number of features to develop, or project risk.

The next set of factors selected by four Web companies were the following:

- Adaptation Effort of Features off the shelf
- Development Effort of New Features
- Effort to Develop User Interface
- Project Risk Factor
- Effort Production testing

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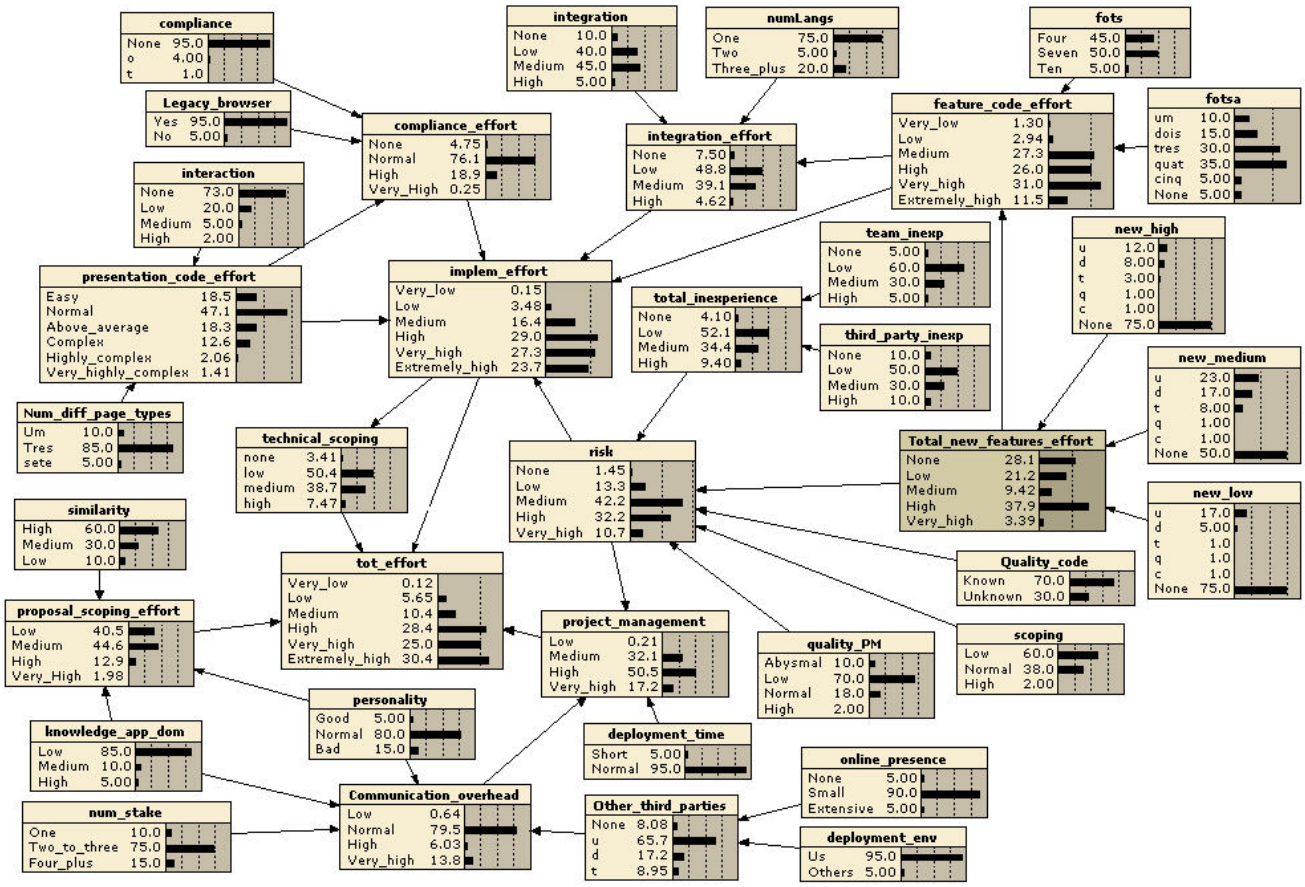

Figure 4. Web Effort Model created for Company C

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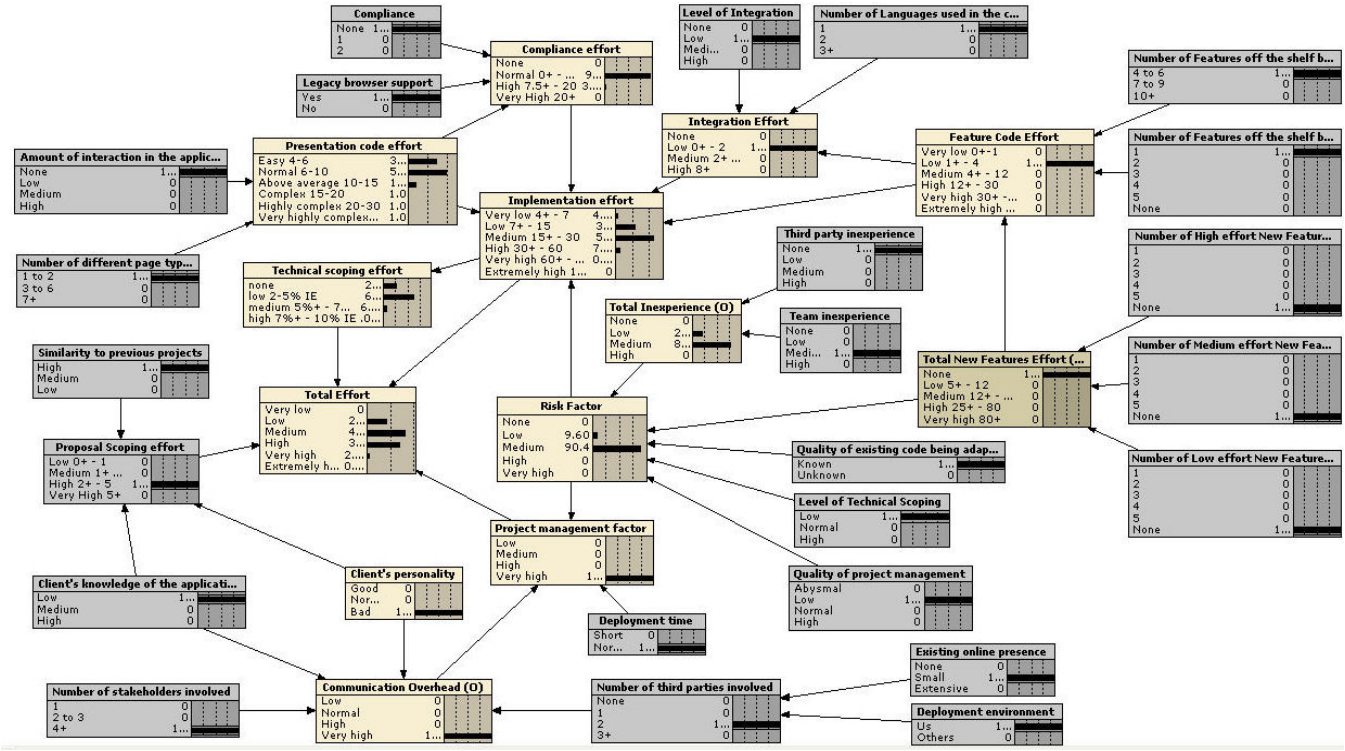

Figure 5. Example of Evidence being entered in the Web Effort Model created for Company C

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TABLE III. COMMON FACTORS AMONGST COMPANIES

Factor	Scale	#Companies
Total Effort	person hours	6
Average Project Team Experience with Technology	years	5
Effort to Program Features	person hours	5
Project Management Effort	person hours	5
Adaptation Effort of Features off the shelf	person hours	4
Development Effort of New Features	person hours	4
Effort to Develop User Interface	person hours	4
Project Risk Factor	UD (e.g. low, medium, high)	4
Effort Production testing	person hours	4
Client Personality Difficulty	UD (e.g. Low, Medium, High; good, normal, bad)	3
Effort producing Animations using Software	person hours	3
Effort to Implement the Web application	person hours	3
Effort to Produce Requirements Documentation	person hours	3
Effort to Produce Template Mock-up	person hours	3
Effort to Produce Web Pages	person hours	3
How much Technical planning	UD (e.g. low, normal, high)	3
Number of Features off the shelf	Integer	3
Number of Features off the shelf Adapted	Integer	3
Number of New Web Pages	integer	3
Effort Post-release testing	person hours	3
Deployment time	UD (e.g. short, normal)	3
Development team size	integer	3
Effort Images Manipulation	person hours	3
Effort to Integrate New and Reused Features	person hours	3
Quality of Project Management	UD (e.g. abysmal, low, normal, high)	3
Technical planning effort	person hours	3
Level of Integration between Features	UD (e.g. low, medium, high)	3
Effort to Design Content	person hours	3
Effort programming Animations	person hours	2
Effort Template Look & Feel	person hours	2
Web Company's Hosting Control	UD (e.g. client in-house, shared, dedicated, in-house)	2
Is Development Process Documented?	Yes/No	2
Number of Features requiring High effort to create	Integer	2
Number of Features requiring Low effort to create	Integer	2
Number of Features requiring Medium effort to create	Integer	2
Number of Key Client's people	Integer	2
Number of Reused Web Pages	integer	2
Number of third parties involved	Integer (e.g. sub-contractors, printing, SMS gateways, hosting providers, domain registration, payment providers)	2
Quality of In-house Existing Code	UD (e.g. low, normal, high)	2
Quality of Third Party Deliverables	UD (e.g. low, high)	2
Type of Project	UD (e.g. New, Enhancement)	2
Number of Natural Languages Used	integer	2
Total third party inexperience	UD (e.g. low, medium, high)	2
unknown technology risk	UD (boolean)	2
Amount of text per Application	UD (e.g. low, medium, High)	1
Client Application Domain Literacy	UD (e.g. low, medium, High)	1
Client's Existing online presence	UD (e.g. small, extensive, none)	1
Development Process Model	UD (e.g. conventional, waterfall, extreme)	1
Number of Images requiring High effort to manipulate	integer	1
Number of Images requiring Low effort to manipulate	integer	1
Number of Images requiring Medium effort to manipulate	integer	1
Number of Web Page Templates	integer	1
Level of Usability	UD (e.g. low, medium, high)	1
Similarity to Previous Projects	UD (similarity of domain/functionality/design; e.g.: low, medium, high)	1
Legacy browser support	UD (e.g. yes, no)	1
Effort to Implement Accessibility	person hours	1

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Forum Feature	UD (boolean)	1
User sign up feature	UD (boolean)	1
Auction system feature	UD (boolean)	1
types of Listing features	UD catagories	1
Gallery feature (number of controls)	UD (number of widgets)	1
shopping cart feature	UD (boolean)	1
event calander feature	UD (boolean)	1
Number of Blogs	Integer	1
Number of Poll	integer	1
Mailling List feature	UD (boolean)	1
Effort to produce user documentation	person hours	1
Tight schedule	Bollean	1
Template design uniqueness	UD (e.g. Template Standard, Template High, Custom-medium, Custom-high	1
Effort to Implement the Template	person hours	1

Except for Project Risk Factor and Average Project Team Experience with Technology, all the remaining factors related to the effort to accomplish certain tasks, such as adapting or developing a new feature, testing and interface design. Note that these factors are very much related to more dynamic Web applications, which offer a large set of features (this requiring more detailed testing). It is also interesting that the effort to develop the user interface was also chosen by four of the six companies. Nowadays, with the plethora of Web technologies and possibilities available, good interface design and usability can also add very much so to a company's competitive advantage on the global market.

All six BNs presented several effort-related factors as predictors of Total effort. We believe that this occurred because all models also reflected the specific effort estimation workflows employed by each of the participating companies. For example, when you look at Figure 4, if you target at the leaf factors first, and then follow the causal relationships until you reach Total Effort, it is clear that the paths followed also correspond to the process used to obtain bottom up effort estimates, which are then combined into Total Effort.

V. LESSONS LEARNT

This research has provided numerous lessons, as follows:

First: engaging with industry. At the start of this research, in order to reach out to industry, we invited the local NZ IT industry to attend a seminar about Web effort estimation and how to improve their estimates. The seminar provided an introduction to using expert-based BNs, their value as estimation tools, and their capability for running "what-if" scenarios. Many of the participating companies saw the immediate value in such an approach, in particular because it enabled the very close and fundamental participation of in-house domain experts while building and validating the company-specific model. Several companies sign up to collaborate.

Second: Time constraints. Depending on the complexity of the BN model, the elicitation of probabilities can be very time consuming and last several months. At the start of the research project we did not have the means to provide the automatic generation of probabilities, so all probabilities had to be elicited manually based on expert knowledge. This was a drawback, given that the time needed to develop a full-fledged model became prohibitive to some companies. As a consequence, out of the initial set of 10 NZ participating companies, only five remained until the full completion of their BN models. Motivated by this issue, we also investigated mechanisms to enable the automatic generation of probability tables. A solution was devised and used with two NZ companies that participated more recently in this research. This solution comprised the comparison between different probability generation algorithms and expert-driven probabilities. Therefore we needed first to have completed the probability elicitation and validation of several BN models so to have a basis for comparison with the proposed algorithms. A tool was implemented as a result of this work; further details can be found in [2][3].

Third: Value for a company. Except for the company in Brazil, the other participating companies were contacted for post-mortem interviews. The main points highlighted were the following:

- The elicitation process enabled experts to think deeply about their effort estimation process and the factors taken into account during that process, which in itself was considered advantageous to the companies. This has been pointed out to us by all the DEs interviewed.
- Once a BN model was validated, DEs started to use their model not only for obtaining better estimates than the ones previously prepared by subjective means, but also sometimes as means to guide their requirements elicitation meetings with prospective clients. They focused their questions targeting at obtaining evidence to be entered in the model as the requirements meetings took place; by doing so they basically had effort estimates that were practically ready to use for costing the projects, even when meetings with clients had short durations. Such change in approach proved to be extremely beneficial to the companies given that all estimates provided using the models turned out to be more accurate on average than the ones previously obtained by subjective means.
- Clients were not presented the models due to their complexity; however by entering evidence while a requirements elicitation meeting took place enabled the DEs to optimize their elicitation process by being focused and factor-driven.

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- One of the participating companies, the largest company in total number of employees, and also the one that built the largest BN model, provided the following feedback: The DEs who participated in the causal structure and probabilities' elicitation changed completely their approach to estimating effort as follows: they presented the BN model to all of their development teams, and asked that from that point onwards every estimate for any task should be based on the factors that had been elicited. This means that an entire team started to use the factors that have been elicited, as well as the BN model, as basis for their effort & risk estimation sessions. In addition, the DEs presented the model at a meeting with other company branches, so to detail how the Auckland branch was estimating effort and risk for their healthcare projects. The other branches were so impressed, in particular the one from the US, that they increased the number of Healthcare software projects outsourced to the NZ Branch, as they recognised the benefits of using a model that represented factors and uncertainties. Overall, such change in approach provided extremely beneficial to the company.

All the companies remained positive and very satisfied with the results. We believe that the successful development of these six Web effort BN models was greatly influenced by the commitment of the participating companies, and also by the DEs' experience estimating effort.

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REFERENCES

[1] Baker, S., and Mendes, E.(2010) Aggregating Expert-driven causal maps for Web Effort Estimation, Proceedings of the International Conference on Advanced Software Engineering & Its Applications (ASEA), Volume 117, pp. 264-282.

[2] Baker, S., Mendes, E. (2010) Assessing the Weighted Sum Algorithm for Automatic Generation of Probabilities in Bayesian Networks, Proceedings of the International Conference on Information and Automation 2010 (ICIA 2010), pp. 867-873, DOI: 10.1109/ICINFA.2010.5512447

[3] Baker, S., Mendes, E. (2010) Evaluating the Weighted Sum Algorithm for Estimating Conditional Probabilities in Bayesian Networks, Proceedings of the Software Engineering and Knowledge Engineering Conference (SEKE 2010), pp. 319-324.

[4] Das, B.: Generating Conditional Probabilities for Bayesian Networks: Easing the Knowledge Acquisition Problem, arxiv.org/pdf/cs/0411034v1 (accessed in 2008). (2004)

[5] Druzdzel, M.J., and van der Gaag, L.C.: Building Probabilistic Networks: Where Do the Numbers Come From?, IEEE Trans. on Knowledge and Data Engineering, 12(4), 481-486. (2000)

[6] Fenton, N., Marsh, W., Neil, M., Cates, P., Forey, S. and Tailor, M.: Making Resource Decisions for Software Projects, Proc. ICSE'04, pp. 397-406. (2004)

[7] Jensen, F. V.: An introduction to Bayesian networks. UCL Press, London. (1996)

[8] Mahoney, S.M., and Laskey, K.B.: Network Engineering for Complex Belief Networks, Proc. Twelfth Annual Conference on Uncertainty in Artificial Intelligence, pp. 389-396. (1996).

[9] Mendes, E.: (2007)Predicting Web Development Effort Using a Bayesian Network, Proceedings of EASE'07, pp. 83-93..

[10] Mendes, E.: (2007).The Use of a Bayesian Network for Web Effort Estimation, Proceedings of International Conference on Web Engineering, pp. 90-104, LNCS 4607.

[11] Mendes, E.: (2007).A Comparison of Techniques for Web Effort Estimation, Proceedings of the ACM/IEEE International Symposium on Empirical Software Engineering, pp. 334-343.

[12] Mendes, E.: (2008).The Use of Bayesian Networks for Web Effort Estimation: Further Investigation, Proceedings of ICWE'08, pp. 2-3-216.

[13] Mendes, E.: (2011). Building a Web Effort Estimation model through Knowledge Elicitation, Proceedings of the 13th International Conference on Enterprise Information Systems, pp. 128-135

[14] Mendes, E.: (2011) Improving Project Management of Healthcare Projects through Knowledge Elicitation, Handbook of Research on ICTs for Healthcare and Social Services: Developments and Applications, Isabel Maria Miranda & Maria Manuela Cruz-Cunha (Eds), IGI Global. (Accepted for publication)

[15] Mendes, E.: (2011) Uncertainty-based software effort estimation, IFPUG book, IFPUG (Accepted for Publication).

[16] Mendes, E., and Mosley, N.: (2008)Bayesian Network Models for Web Effort Prediction: a Comparative Study, Transactions on Software Engineering, Vol. 34, Issue: 6, Nov/Dec 2008, pp. 723-737..

[17] Mendes, E., Mosley, N. and Counsell, S.: (2005).The Need for Web Engineering: An Introduction, Web Engineering, Springer-Verlag, Mendes, E. and Mosley, N. (Eds.) ISBN: 3-540-28196-7, pp. 1-28.

[18] Mendes, E., Mosley, N., and Counsell, S.: (2005).Investigating Web Size Metrics for Early Web Cost Estimation, Journal of Systems and Software, 77(2), 157-172.

[19] Mendes, E., Pollino, C., and Mosley, N.: (2009).Building an Expert-based Web Effort Estimation Model using Bayesian Networks, Proceedings of the EASE Conference, pp. 1-10.

[20] Neil, M., Fenton, N., and Nielsen, L.: Building Large-scale bayesian networks, The knowledge Engineering Review, KER, Vol. 15, No. 3, 257-284. (2000).

[21] Pearl J.: Probabilistic Reasoning in Intelligent Systems, Morgan Kaufmann, San Mateo, CA. (1988).

[22] Pendharkar, P.C., Subramanian, G.H., and Rodger, J.A.: A Probabilistic Model for Predicting Software Development Effort, IEEE Trans. Software Eng. Vol. 31, No. 7, 615-624. (2005).

[23] Studer, R., Benjamins, V.R. and Fensel, D.: Knowledge engineering: principles and methods. Data & Knowledge Engineering, vol. 25, 161-197. (1998).

[24] Tang, Z., and B. McCabe: Developing Complete Conditional Probability Tables from Fractional Data for Bayesian Belief Networks, Journal of Computing in Civil Engineering, 21(4), pp. 265-276. (2007).

[25] Woodberry, O., A. Nicholson, K. Korb, and C. Pollino: Parameterising Bayesian Networks, Proc. Australian Conference on Artificial Intelligence, pp. 1101-1107. (2004).

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